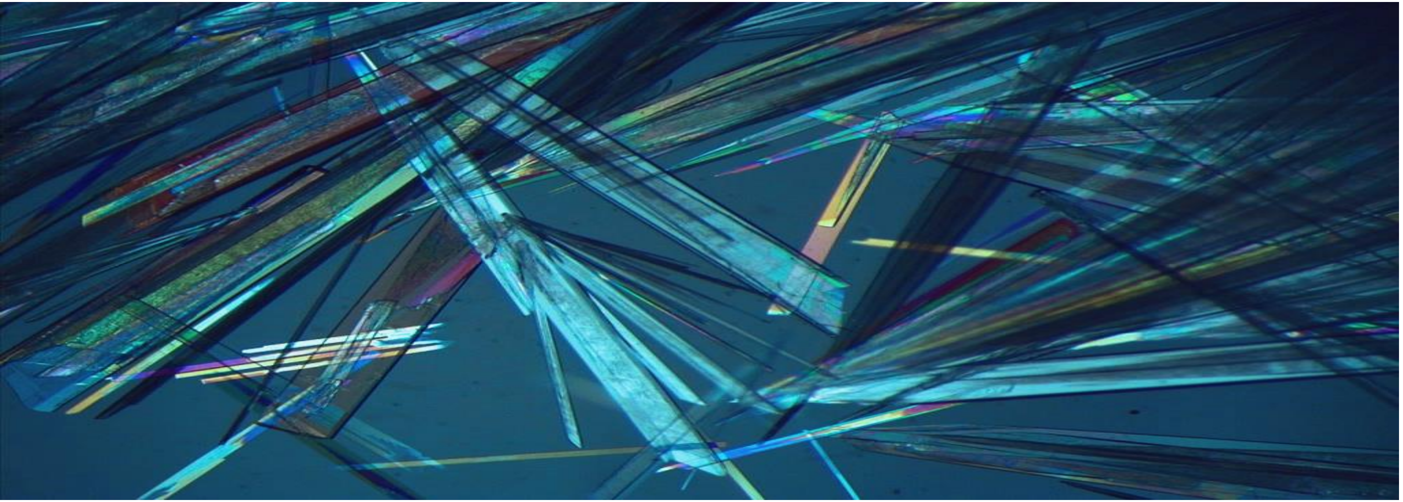


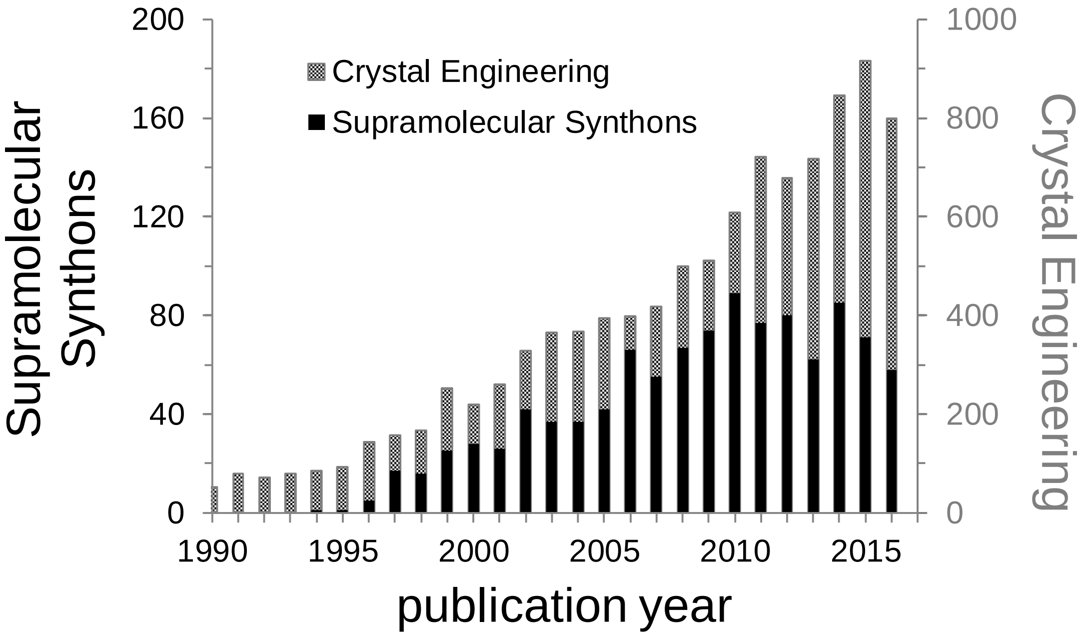
Crystal Engineering im Spezialisierungsmodul Advanced Materials (AdMat)

LEGO mit molekularen Bausteinen

Crystal Engineering: young field with many perspectives

Heinrich Heine
Universität Düsseldorf

A survey of articles on the topic of CE from 1990 to 2016



Is CE Inorganic Chemistry?

Heinrich Heine
Universität Düsseldorf

Supramolecular chemistry

Solid-state chemistry

Crystal Engineering

Material science

Crystallography

Where everything begun?... Supramolecular chemistry

Heinrich Heine Universität Düsseldorf

Molecule – focus of classical chemistry

- ultimate delimiter of all significant properties of a substance.
- only structures and functions involving covalent bond

BUT: some biological systems/phenomena do not involve making or breaking chemical bonds.

INSTEAD: bio systems are usually formed of loose aggregates held together by non-covalent interactions

FURTHER: Some fundamental properties cannot be explained by molecular structure.

Supramolecular chemistry: main principles

Heinrich Heine Universität Düsseldorf

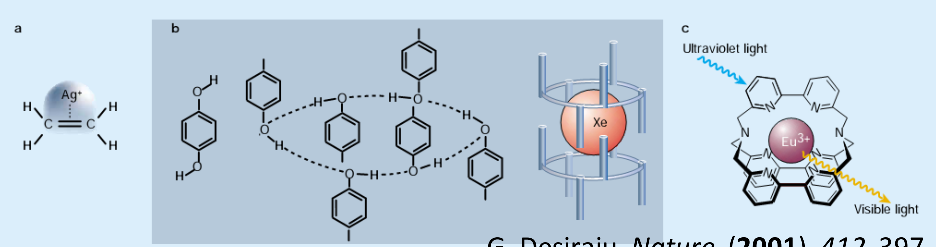
Molecular recognition

* lock-and-key model

* enzyme-substrate interaction (E. Fischer 1894)

* molecules recognise one another through their dissimilarities rather than through their similarities

Chemistry beyond the molecule



G. Desiraju, *Nature*, (**2001**), *412*, 397.

Supramolecular function

* highly specific interactions lead to useful supramolecular functions

$\Rightarrow$ Interest - stable compounds that do not involve covalent bonds

Supramolecular chemistry is a chemistry of the intermolecular bond

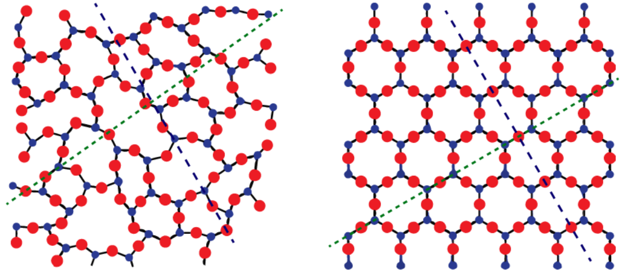
J.-M. Lehn, 1969

Crystal Engineering (CE) –

understanding of intermolecular interactions in the context of crystal packing

why do we study CE? **Aim**: design of new solid materials with desired chemical and physical properties

Crystal is an <u>ordered</u> and <u>periodic</u> arrangement of molecules/atoms



design of solids with desired properties ↔ means design crystals where molecule are organised in certain ways.

What do we do?

→ make crystals by using interactions

How do we do it?

→ synthetic strategies

Why do we do it?

→ properties

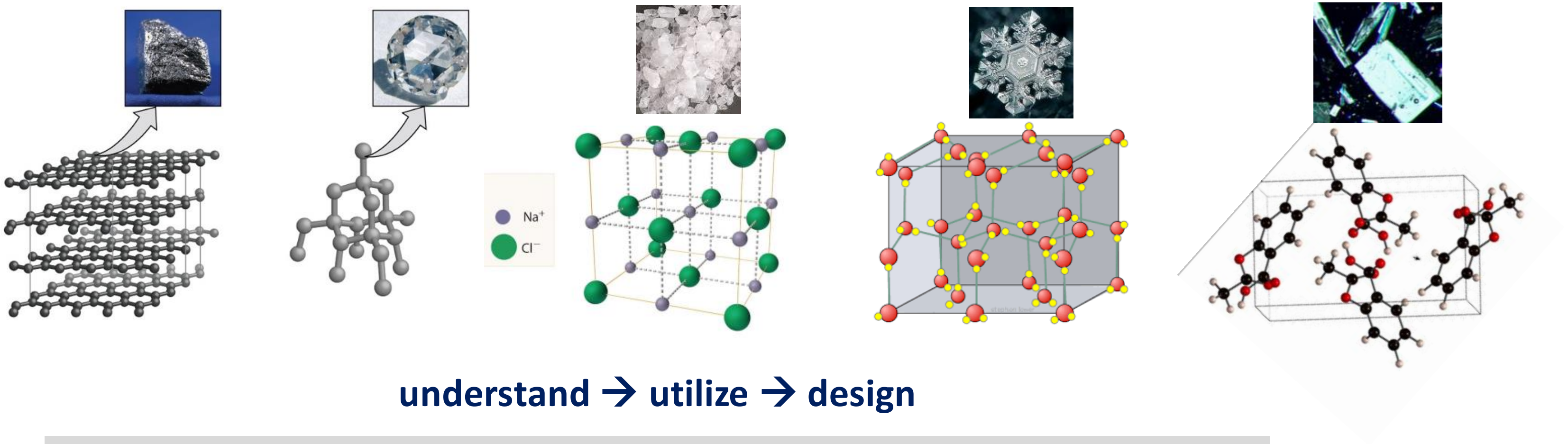
<u>– operations in CE –</u>

- Determination of crystal structures
- Understanding or analysis of crystal structures
- Use of understanding in trying to design a crystal structure
- Actual crystallization experiment
- Realization of pre-desired crystal property

What is a molecular crystal?

Molecular crystal is a **collection of molecules**, held internally by rather strong chemical bonds and associated with each other by weaker intermolecular interactions.

➢ molecular crystal is a crystalline form of any chemical substance that exists as molecule!



understand → utilize → design

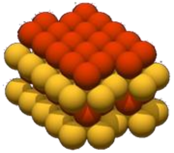
Fundamental question of modern CE:
Given a molecular structure of a compound, what is its crystal structure?

Understanding tools

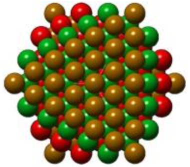
for aggregation phenomena of molecules

Close packing principle

molecules pack in a crystal so that any molecule is surrounded by a maximum number of close neighbors, generally 12.

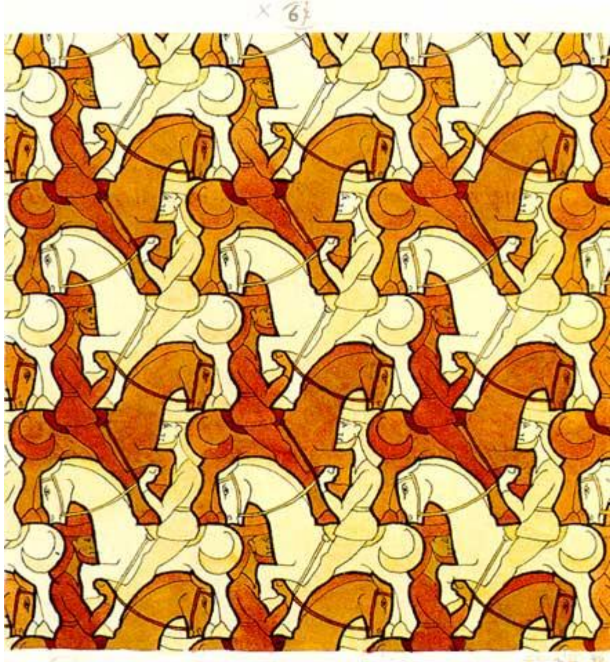


(hcp)



(ccp)

- minimization of the potential energy of the system
- isotropic attractive and repulsive forces between molecules
- highest possible number of stabilizing interactions



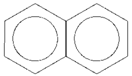
M.C. Escher's “Horsemen”

Understanding tools: close packing

Condensed aromatic hydrocarbons C_mH_n

naphthalene

$C_{10}H_8$

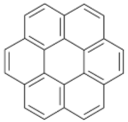


m/n < 1.5

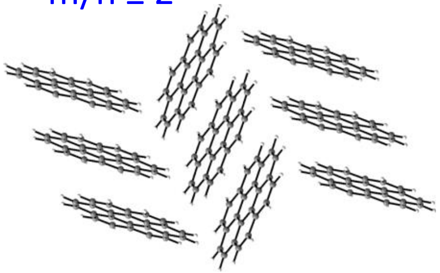
- hydrogen rich molecules
- angle ~40-50°
- herringbone geometry
- C-H··· π interactions

coronene

$C_{24}H_{12}$



m/n ≥ 2

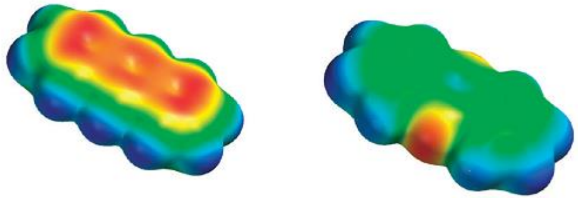


- carbon rich molecules (m/n ≥ 2)
- angle ~90°
- tendency toward π···π stacking

➢ C/H-stoichiometry is a critical parameter

Acenes vs N-Heteroacenes: The Effect of N-Substitution on the Structural Features of Crystals of Polycyclic Aromatic Hydrocarbons

Kenneth E. Maly



anthracene phenazine

calculated electrostatic potential map

- importance of C···H contacts is reduced
- C-H···N interactions in phenazine
- increased tendency to π···π stacking

➢ Heteroatom impact on crystal packing

K.E. Maly, *Cryst. Growth Des.*, **2011**, *11*, 5628-5633.

Understanding tools

for aggregation phenomena of molecules

Intermolecular interactions

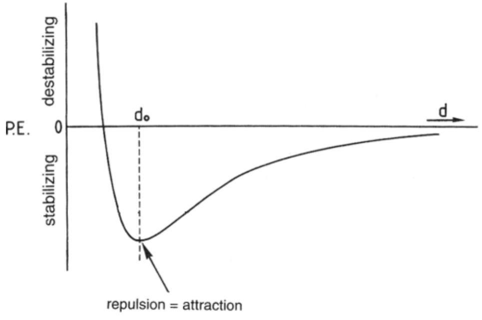
(between 2 atoms) ≤ ∑ of their van der Waals radii.

Strength > **weak and strong** stabilizing & destabilizing **energies**

Distance > electrostatic and dispersive

Directionality > isotropic and anisotropic

Potential energy diagram for an intermolecular interaction



d0 - equilibrium distance
→ lowest energy
→ repulsion = attraction

distances d > d0 (and some shorter)
→ stabilizing E < 0
→ attractive forces

very short distances d < d0
→ destabilizing E > 0
→ repulsive forces

stabilizing & destabilizing energies describe intermolecular interactions

Properties of intermolecular interactions

- **Strength**

  short/long & weak/strong

  distance & energy

- majority of covalent bonds: 300 - 500 kJ/mol
- majority of hydrogen bonds: 4 - 60 kJ/mol

Bond/ Interaction	Bond length (Å)
C-H	
C-C	
C=C C≡C	
C-O	
C=O	
O-H	
N-H	
O···H	

Bond/ Interaction	Bond energy (kJ/mol)
O-H	
C-H	
C-C	
C-N	
S-S	
Hydrogen bond	
Van der Waals (vdW)	~

Properties of intermolecular interactions

➢ Strength

weak and strong

- majority of covalent bonds: 300 - 500 kJ/mol
- majority of hydrogen bonds: 4 - 60 kJ/mol

➢ Directionality

isotropic and anisotropic

isotropic interactions
non-directional
simple level of molecular recognition

→ responsible for supramolecular arrangements
→ close packing of the molecules
→ include vdW → vdW forces: attractive or repulsive (between all atoms and molecules)

anisotropic interactions
directional
higher level of molecular recognition (lock-and-key principle)

→ determine intermolecular orientation and functions
→ involve partially charges atoms (N, O, Cl, P, S)
→ have electrostatic character – dipole-dipole interaction

→ master key – hydrogen bond – fits to different locks

Some important intermolecular interactions in CE

- Hydrogen bonds (HB)
- Halogen bonding (XB)
- $\pi \cdots \pi$ interactions (stacking)

Definition & Properties of H-bonds

Heinrich Heine Universität Düsseldorf

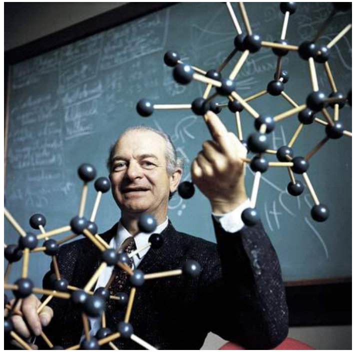
Master key of molecular recognition, **“the” toolkit** of CE

An attractive interaction **D–H···A–Z** between a hydrogen atom from molecular fragment D−H in which D is more electronegative than H, and an atom / group of atoms in the same or a different molecule. *Current IUPAC definition (2011)*

H is more electropositive than D (hydrogen bond donor) and A (HB acceptor)
→ **electrostatic character**

HB Properties

- Strength & length
- Directionality
- Cooperativity
- Hierachy



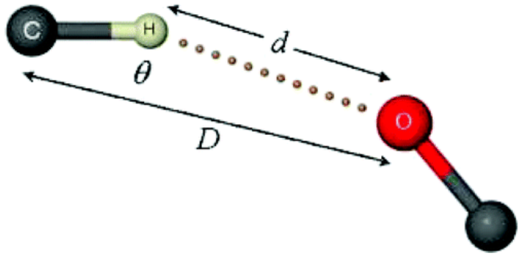
Linus Pauling

Gautam Desiraju

E. Arunan, G. R. Desiraju, R. A. Kleinet al, *Pure Appl. Chem.* **2011**, *83*, 1637.

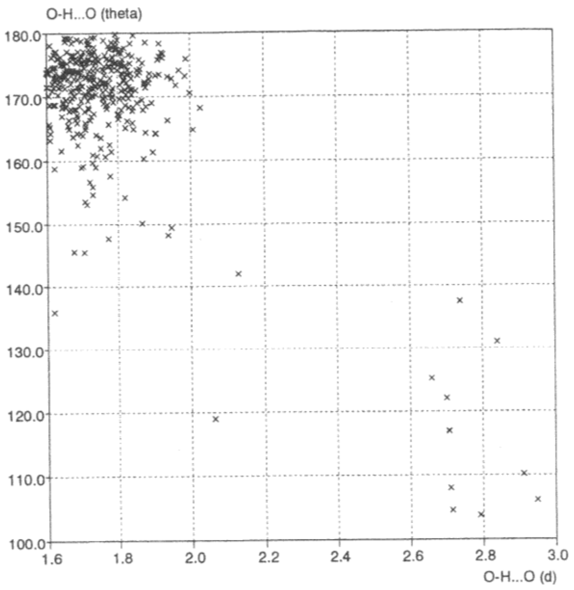
Hydrogen bond, X–H···Y–Z

Strength	Example	X···Y (*D*, Å)	H···Y (*d*, Å)	X–H···Y (*θ*,º)	Bond energy (-kcal/mol)	Effect on crystal packing
very strong X−H ~ H···Y	$[F-H-F]^-$	2.2–2.5	1.2–1.5	175–180	15-40	strong
strong X−H < H···Y	O−H···O–H O−H···N–H N–H···O=C	2.6–3.0 2.6–3.0 2.8–3.0	1.6–2.2 1.7–2.3 1.8–2.3	145–180 150–180 150–180	4-15	distinctive
weak X−H << H···Y	C−H···O O−H···π	3.0–4.0	2.0–3.0	110–180 90–180	< 4	variable



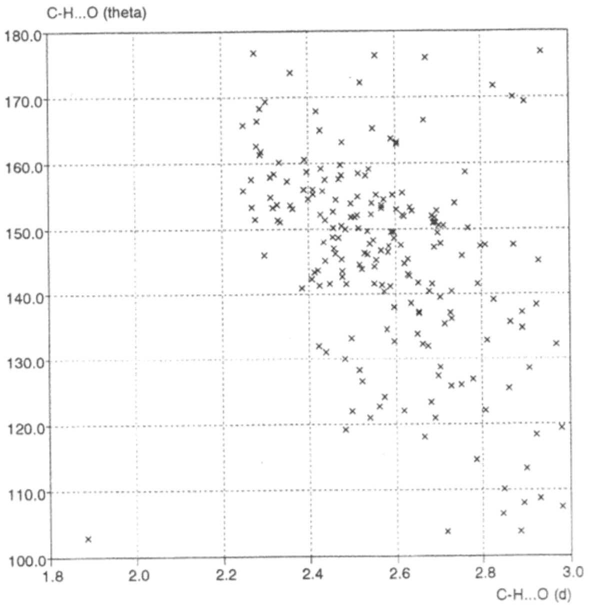
Cambridge Structural Database (CSD) analysis

strong HBs



O-H···O bonds in carboxylic acid dimers

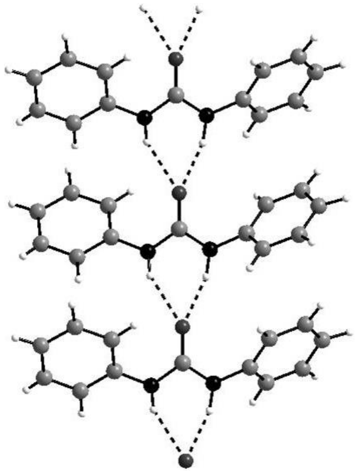
weak HBs



C-H···O bonds in α,β-unsaturated carbonyl compounds

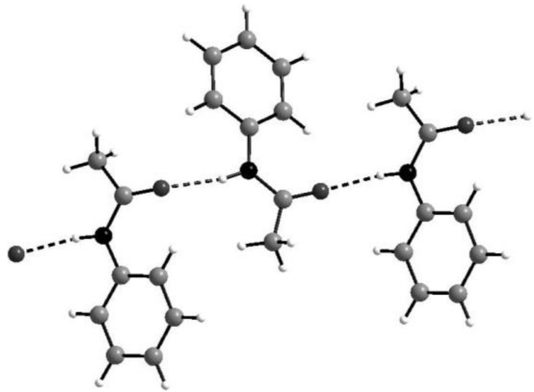
Angular properties of hydrogen bonds

Diphenylurea



A bifurcated hydrogen bond, or
three-centered interaction
- two equiv. donors, one acceptor

Acetanilide



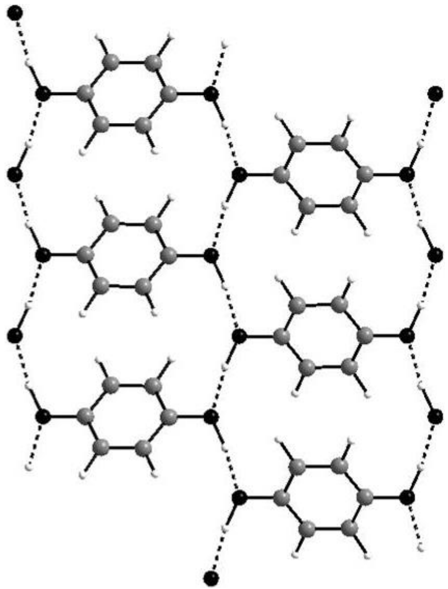
Linear N–H···O=C bonds
connect secondary amide molecules in a one-dimensional array

- two-centered interaction
- linear angle D-H···A = 180 °

Cooperativity of HBs

e.g. alcohol **R-OH** (γ-hydroquinone):

acidity of H-atom in O–H···O–H dimer **>** in an isolated O–H group



HBs are more stable when involved in dimers, trimers, chains, infinite 2D- and 3D structures!

Why?

Polarizability & charge transfer effects:

total binding energy of all HB in an aggregate **> energy sum of the individual** HBs

Hierarchy of HBs

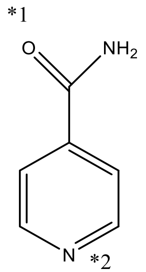
HB donors: OH, NH and NH_2

HB acceptors: **O**H, C=**O**, **N**H_2, -**N**= and –**O**-

In case of several HB donors and acceptors they pair up along their strength

⇒ the **best H-bond** is formed by the **best H-bond donor** and H-bond **acceptor**

⇒ the **second best HB**: by the **second best HB donor** and the second HB **acceptor**



Experiment:
mix isonicotinamide + two different aromatic acids (1:1:1)

*1:
*2:

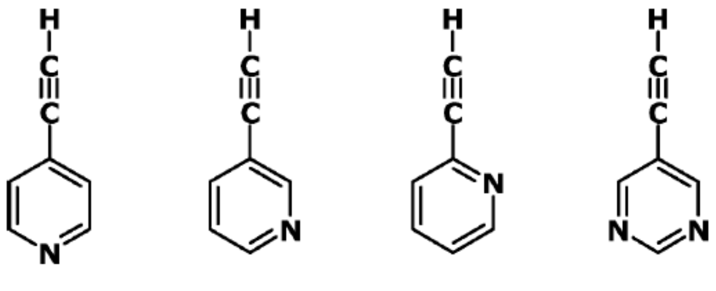
Hierarchy is a characteristic of kinetic (fast) crystallization

CE with weak HBs

Crystal Engineering with ≡C−H···N and =C−H···N Hydrogen Bonds

Venkat R. Thalladi,*,† Marta Dabros,† Annette Gehrke,‡ Hans-Christoph Weiss,‡ and Roland Boese*,‡

Crystal Growth & Design, Vol. 7, No. 4, 2007



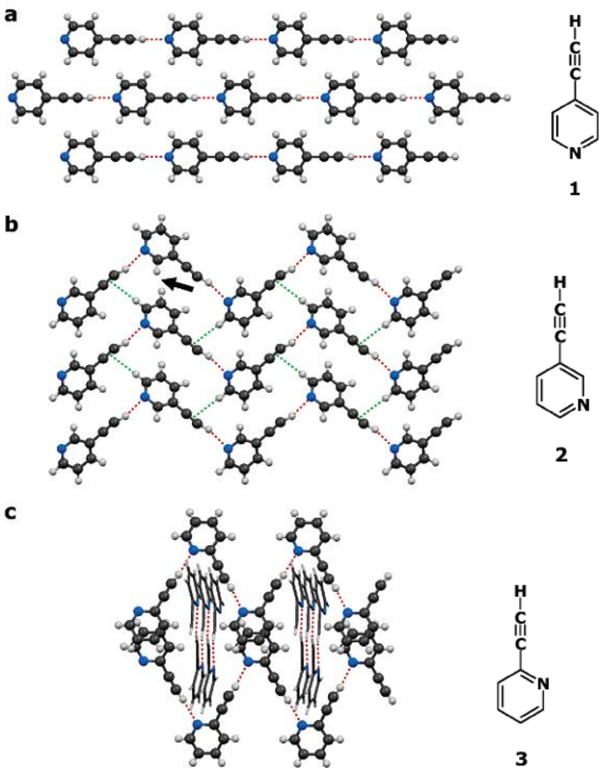
1 2 3 4

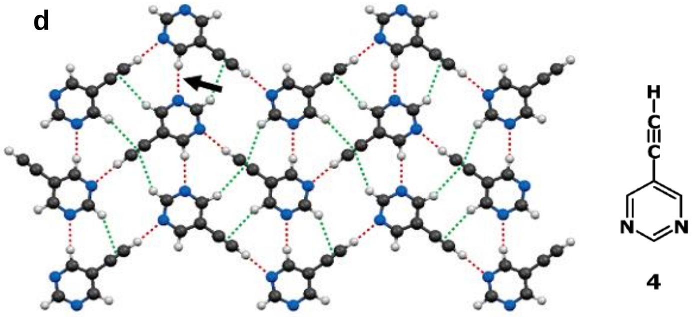
Ethynyl-pyridines and -pyrimidine

- assume that ≡C-H / =C-H groups form short & linear hydrogen bonds

- use weak HBs to generate networks with predictable topologies

Crystal Engineering with weak ≡C-H···N HBs





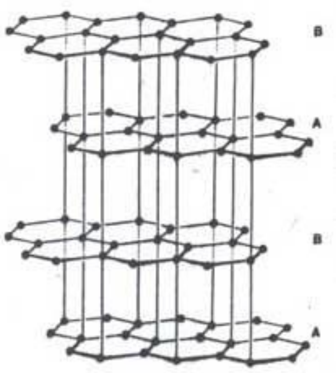
- straight (**1**) and zigzag (**2**,**3**,**4**) chain patterns formed by ≡C-H···N hydrogen bonds

- additional cross-stitching of zigzag chains (**4**) by =C-H···N hydrogen bonds
  → 2D network

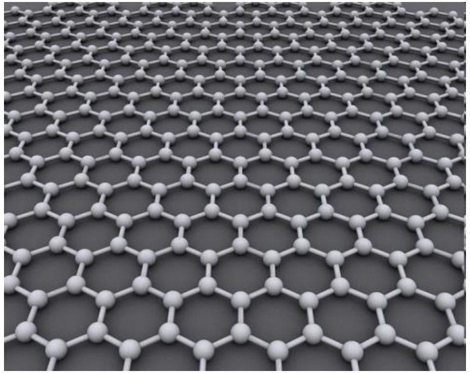
V. R. Thalladi et. al. *Crystal Growth & Design*, **2007**, *7*, 599.

Modulation of analoques of unknown network

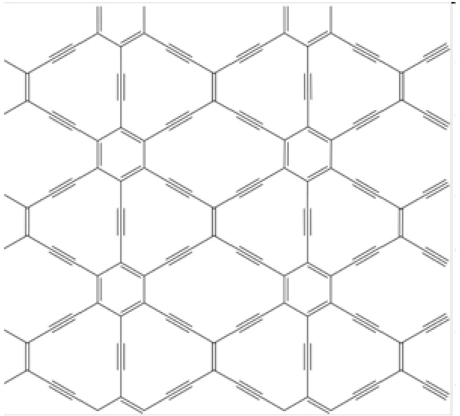
Graphite



Graphene



Graphyne



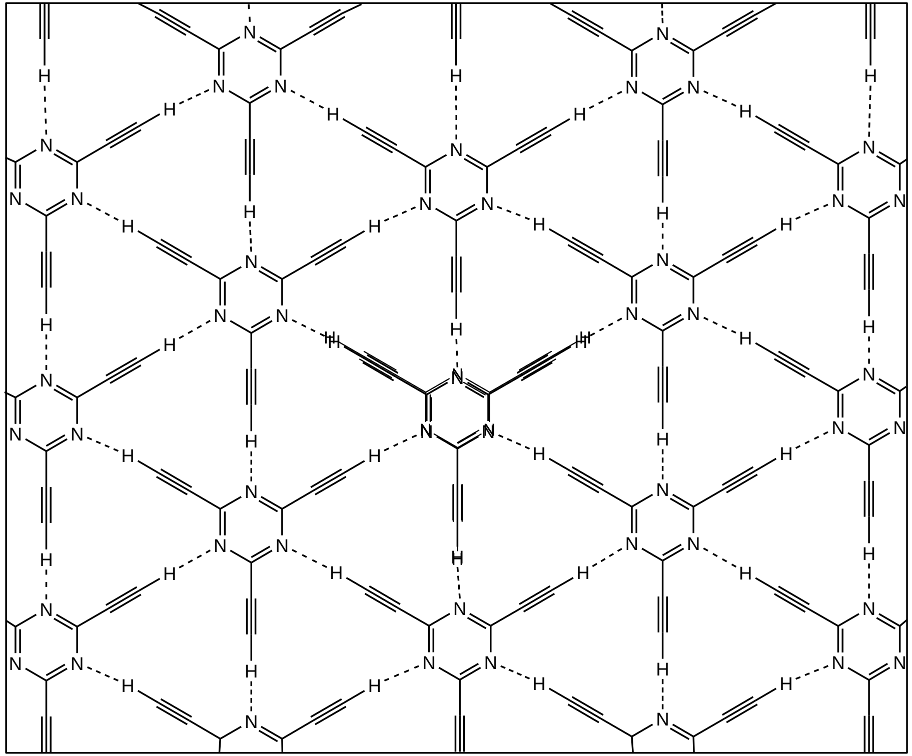
Graphene, a layer of graphite just one atom thick, isn't called a wonder material for nothing. The subject of the 2010 Nobel Prize in physics, it is famed for its superlative mechanical and electronic properties. Yet new computer simulations suggest that the electronic properties of a little-known sister material of graphene—graphyne—may in some ways be better.

The simulations show that graphyne's conduction electrons should travel extremely fast—as they do in graphene—but in only one direction. That property could help researchers design faster transistors and other electronic components that process one-way current, says one of the study's authors, theoretical chemist Andreas Görling of the University of Erlangen-Nuremberg in Germany. "If your material already conducts in one direction, you have a head start," he says.

Supramolecular graphyne:

robust 2D hydrogen-bonded network structure with

suitable model: 2,4,6-triethynyl-1,3,5-triazine



- hexagonal array
- graphyne-like network
- 2D layers
- parallel stacking of layers
- weak C-H···N H-bonds
- <(C-H···N) = 180°
- all N and H atoms involved in the synthon formation

M. Ohkita, *Chem. Comm.* (**2002**) 3054

Building blocks for optoelectronic devices

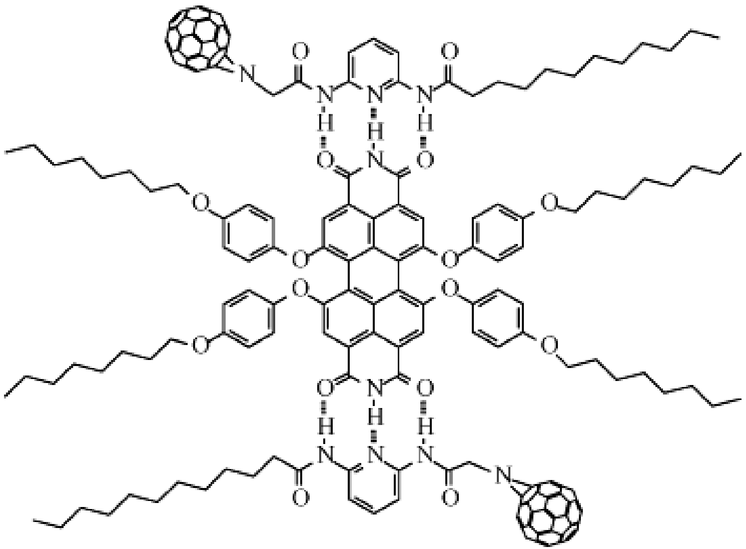
multiple hydrogen bonds for photovoltaic

Superstructure

Self-assembly of C_{60} fullerene derivative with PERY (perylene bisimide)

multiple HBs

DAD sequence



desired construction of functionalised fullerene-based architecture

Yang Liu, *J. Phys. Chem. B*, (**2004**), *108*, 6256.

Halogen Bonding (XB)…

SCIENTIFIC CONFERENCES

A big hello to halogen bonding

Halogen bonding connects a wide range of subjects — from materials science to structural biology, from computation to crystal engineering, and from synthesis to spectroscopy. The 1st International Symposium on Halogen Bonding explored the state of the art in this fast-growing field of research.

Mate Erdelyi

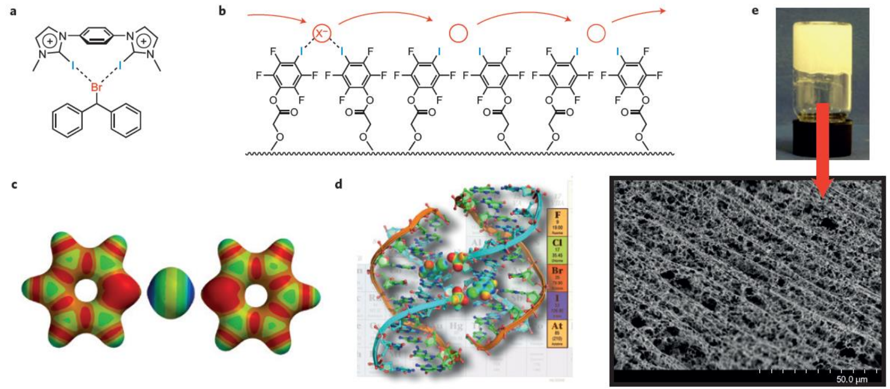


Figure 1 | Some highlights from the ISXB-1 meeting. **a**, A bidentate cationic halogen-bond donor designed for promoting halogen abstraction, for example in the solvolysis of benzhydril bromide. **b**, A synthetic ion channel that operates with halogen bonds. **c**, An iodonium ion is capable of forming two halogen bonds simultaneously, shown in the example of [bis(pyridine)iodine]$^+$. **d**, A DNA junction is an optimal model system to study biomolecular halogen bonds. **e**, An equimolar mixture of bis(pyridyl urea) and diiodotetrafluorobenzene in methanol–water forms a robust hydrogel upon rapid cooling. (Pui Shing Ho and Pierangelo Metrangolo are kindly acknowledged for providing figure parts **d** and **e**, respectively).

NATURE CHEMISTRY | VOL 6 | SEPTEMBER 2014 | www.nature.com/naturechemistry

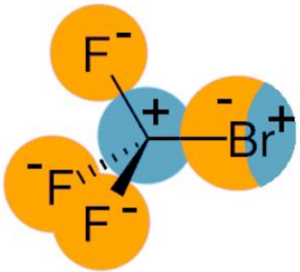
IUPAC Definition

*"A **halogen bond** occurs when there is evidence of a net attractive interaction between an electrophilic region associated with a halogen atom in a molecular entity and a nucleophilic region in another, or the same, molecular entity"*

D-X···A **D-X···X-D**

(X = I, Br, Cl)
(A = O, N, S)

Nature of halogen bond



XB arise from polarization of C-X bond

$\Rightarrow$ X is polarized positively in the region furthest away from the C atom

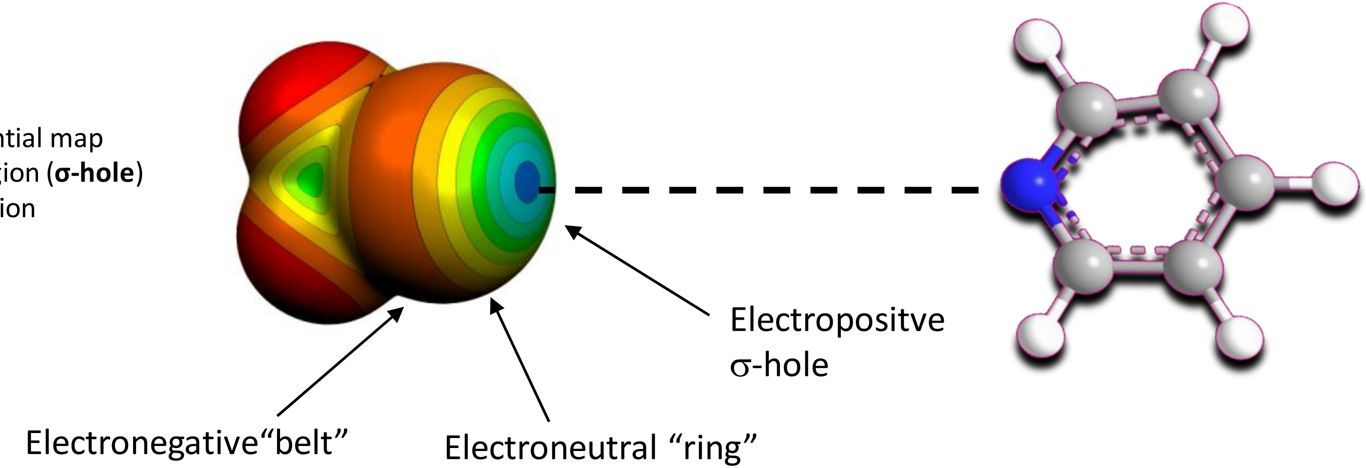
$\Rightarrow$ polarization gradient on C-X: $C^{\delta+}$- $X^{\delta-}$

$\Rightarrow$ Electrostatic contact with a negatively polarized species

$\Rightarrow$ Bond distance shorter than sum of VdW Radii

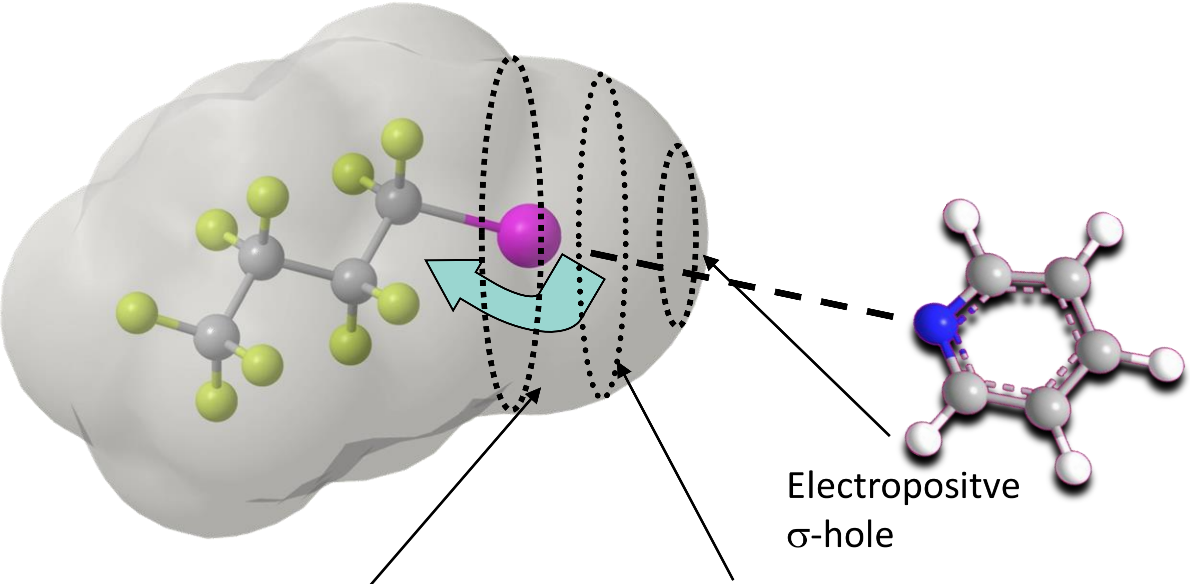
$\Rightarrow$ C-I···N angle approximately 180°

Electrostatic potential map
blue – positive region (**σ-hole**)
red – negative region



The σ-hole

Perfluoroination: enlarges the σ-hole ⇒ interaction strength increases



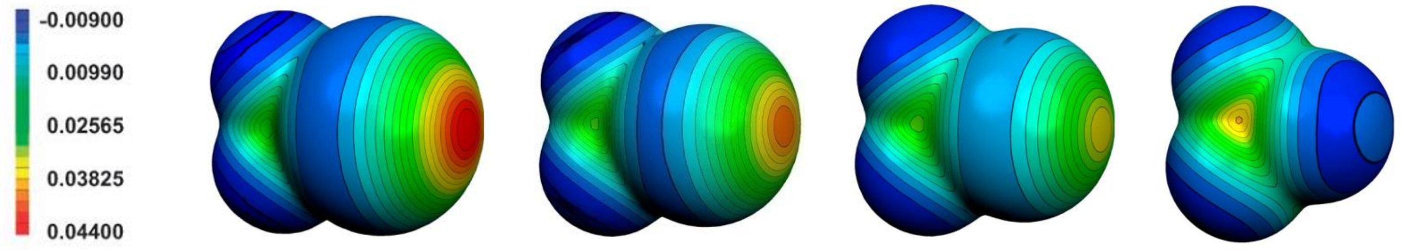
- Polarization of C-X bond

⇒ Electrostatic contact with a negatively polarized species

- Bond distance shorter than sum of VdW Radii
- C-I···N angle approximately 180°

Halogen Bonding (XB)

Perfluoroination: enlarges σ-hole ⇒ interaction strength increases



CF_3I CF_3Br CF_3Cl CF_4

Molecular electrostatic potential of CX_4 - halotrifluoromethane

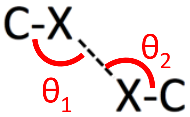
Electrostatic potentials:
red – positive crown – **σ-hole**
blue – negative belt

positive region (σ-hole)

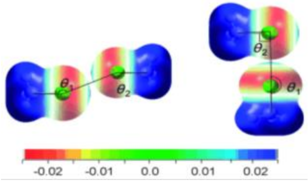
- intermolecular interaction strength I > Br >> Cl > F
- interacts with a N, O, Halogen atom

Geometry of X···X interaction

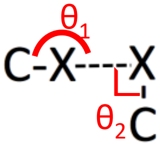
Calculated elektrostatic potentials
- blue – positive polarised,
- red – negatively polarised.



Type I



Type II



Geometry of two energy minimum of chloromethane

- symmetrical
- parallel, electrostatically identical regions
- van der Waals type
- shortest are actually repulsive
- $\theta_1 = \theta_2 \approx 150$
- Torsion angle (C–X···X–C), Φ = 180

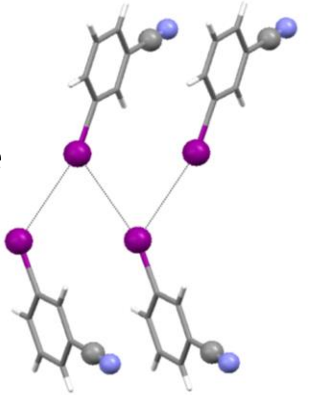
- genuine X-bond
- positive polarisation in the polar region of the halogen atom
- a negative polarisation in its equatorial region
- θ_1 = 180 °und θ_2 = 90 °

K. Merz, V. Vasylyeva, *CrystEngComm*, **2010**, *12*, 3989.

Properties of XB

Linear arrays in ***para***-iodobenzonitrile

meta-iodobenzonitrile
- zig-zag I···I chains



Merz, K., *Crystal Growth & Design*, **2006**, *6*, 1615.

→ Molecular recognition dependant on substitution pattern

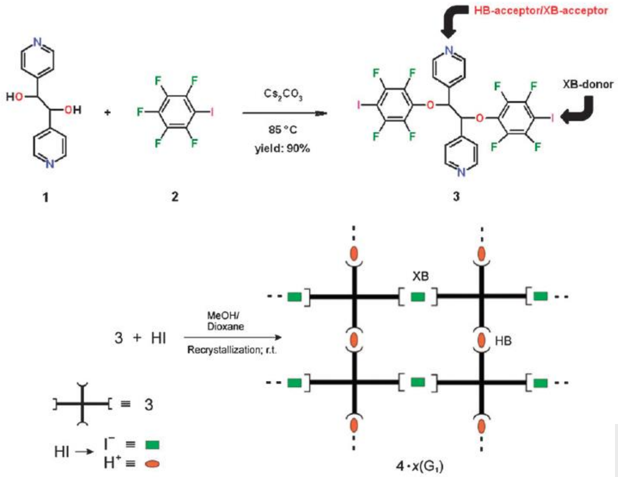
XB Properties

- directional -- σ-hole
- tunable interaction strength -- different depth of the σ-hole
- hydrophobic -- negative and positive sides
- large donor atom size -- tunable: I > Br > Cl

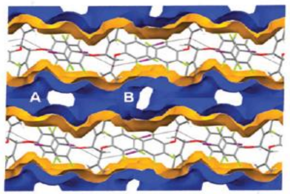
P. Metrangolo, G.Resnati *Halogen Bonding: Applications and Fundamentals*, Springer Verlag, **2008**.

Hydrogen and halogen bonding drive the orthogonal self-assembly of an organic framework possessing 2D channels†

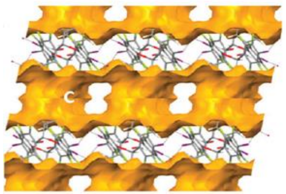
- open framework with 2D channels
- able to host various guest molecules
- single-crystal-to-single-crystal guest exchange



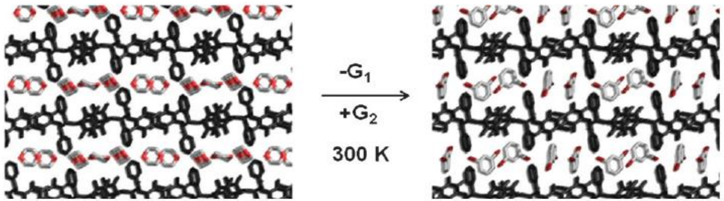
Voids observed after removal of guest molecules.



channels A and B



channel C intersects A and B channels



single-crystal-to-single-crystal guest exchange

XB a promising new tool for the design of open framework materials
→ pre-stage of non-covalently bonded metal-organic frameworks (MOF)

J. Marti-Rujas, L. Colombo, …, G. Resnati, *Chem. Commun.*, **2012**, *48*, 8207. ; contact surface – yellow;,solvent accessible surface – blue

A Cocrystal Strategy to Tune the Luminescent Properties of Stilbene-Type Organic Solid-State Materials**

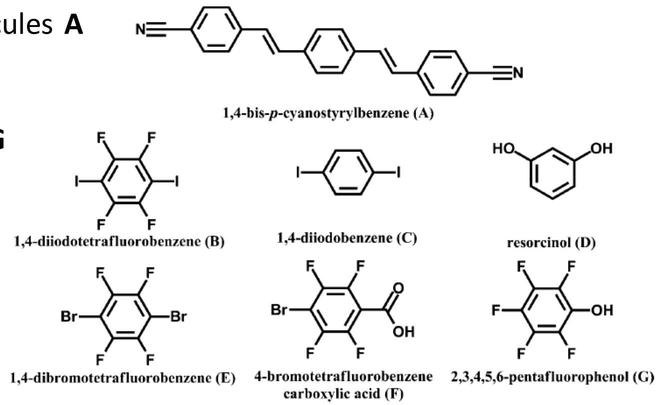
➢ organic solid-state chromophores:

promising optoelectronic applications in the fields of light-emitting diodes, lasers, sensors, and two-photon fluorescent materials.

Challenge: obtain light-emission of an appropriate wavelength

fluorescent molecules **A**

its co-formers **B-G**



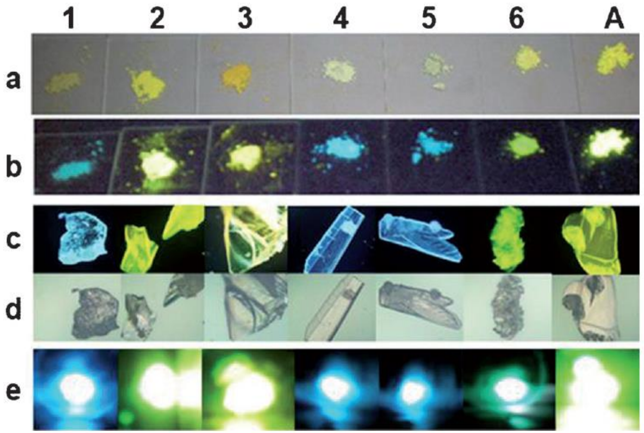


Figure 2. Photographs of solid-state cocrystal samples (from left to right: **1**–**6**) and **A**. a) and b) the powder samples under daylight and UV (365 nm); c) and d) the single crystal samples under UV (365 nm) and daylight observed by fluorescence microscope multiplied by 50-fold; e) two-photon luminescence under 800 nm laser.

Fine-tuning of UV/VIS absorption, luminescence emmision, color, lifetime, quantum yield and structural properties by cocrystal formation

D. Yan et. al, *Angew. Chem. Int. Ed.* , **2011**, *50*, 12483 –12486

XB: new tools and tectons

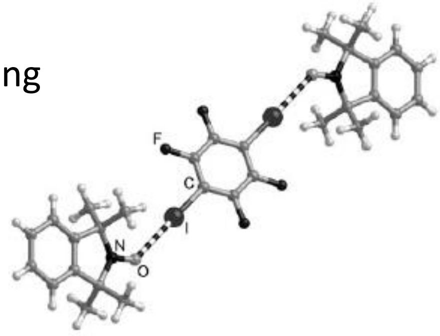
➢ self-assembling organic spin systems

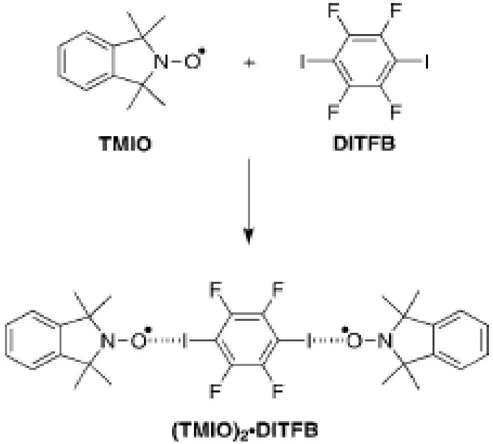
Iodoperfluorocarbons:

→ paramagnetic molecules as building blocks

Potential app.:

- molecular magnetism
- spintronics
- quantum computing





TMIO: tetramethylisoindolinyloxyl
DITFB: diiodotetrafluorobenzene

ionic resonance structure of the radical stabilized by N-O···I interaction

XB in Co-crystals

➢ case study of a preservative

3-Iodo-2-propynyl-*N*-butylcarbamate (**IPBC / Iodocarb**):

antimicrobial product used as preservative, fungicide and algaecide.

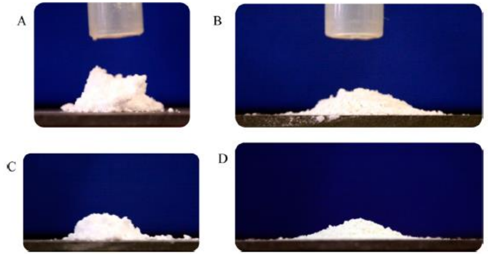
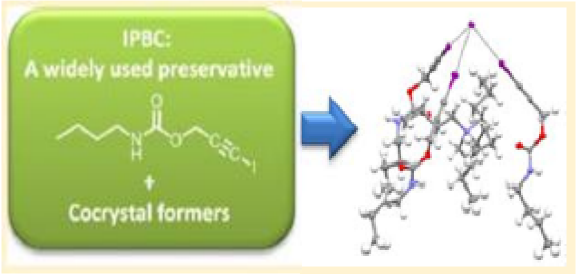
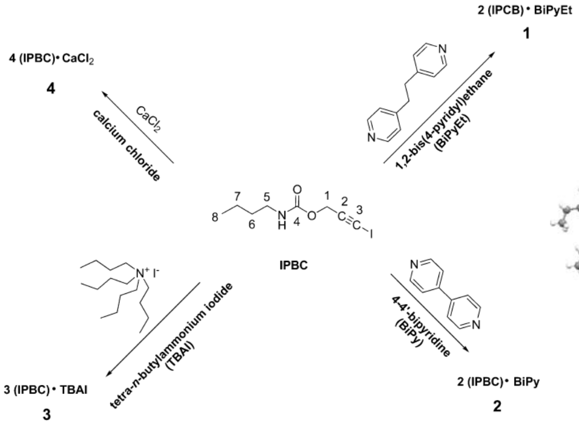


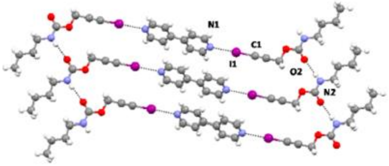
Figure 9. Pictures of cones of IPBC (A, C) and cocrystal 4 (B, D) powders, taken after flowing the powders through the funnel from 25 mm (A, B) and 50 mm (C, D) heights. The cylindrical shape of IPBC cones indicates clearly the high cohesion of the powders, while the flat cone shape of the cocrystal 4 indicates improved powder flow properties.

Problems:
- Diffticult to obtain in pure form
- Hard to handle in industrial products and processes
- Sticky and clumpy

Idea:
- co-crystals







M. Baldrighi et al., *Mol. Pharmaceuticals*, **2013**, *10*, 1760

Fluorine substituent effect

Fluorine substituent effect

1. on physico-chemical properties ?
2. on interactions ?

Special properties of F:
- high electronegativity
- relatively small size
- very low polarisability

Perfluorocarbons

- Nearly ideal liquids.
- High compressibility.
- High vapor pressures.
- Poor solvents for organics.
  - non-polar molecules;
  - low polarizability;
  - small inter- and intra-molecular forces.

Hydrofluorocarbons

- Polar molecules.
- Appreciable inter- and intra-molecular interactions

Fluorine substituent effect

on physico-chemical properties of **fluorocarbons**:
only some tendencies

- Solubility (the least and the most polar solvents)
- Lipophilicity (increases in aromatic systems)
- Inductive effect (electron withdrawing)
- Acidity and basicity (increases HB acidity, decreases basicity)
- Steric effect (depends on compound and fluorination pattern)

Fluorine substituent effect

On interactions?

Fluorine: no σ-hole → no XB formation

What about HB?

F atom should be a stronger H-bond-acceptor than O or N

- correct for HF_2^-
- not correct for organic fluorinated compounds (C-F)

Table 1. Calculated hydrogen bond energies (kcal mol^{-1}) in some gas-phase dimers.[a]

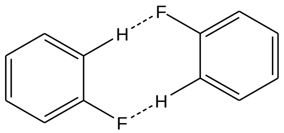
Dimer	Energy
$[F{-}H{-}F]^-$	39
$[H_2O{-}H{-}OH_2]^+$	33
$[H_3N{-}H{-}NH_3]^+$	24
$[HO{-}H{-}OH]^-$	23
$NH_4^+ \cdots OH_2$	19
$NH_4^+ \cdots Bz$	17
$HOH \cdots Cl^-$	13.5
$O{=}C{-}OH \cdots O{=}C{-}OH$	7.4
$HOH \cdots OH_2$	4.7; 5.0
$N{\equiv}C{-}H \cdots OH_2$	3.8
$HOH \cdots Bz$	3.2
$F_3C{-}H \cdots OH_2$	3.1
$Me{-}OH \cdots Bz$	2.8
$F_2HC{-}H \cdots OH_2$	2.1; 2.5
$NH_3 \cdots Bz$	2.2
$HC{\equiv}CH \cdots OH_2$	2.2
$CH_4 \cdots Bz$	1.4
$FH_2C{-}H \cdots OH_2$	1.3
$HC{\equiv}CH \cdots C{\equiv}CH^-$	1.2
$HSH \cdots SH_2$	1.1
$H_2C{=}CH_2 \cdots OH_2$	1.0
$CH_4 \cdots OH_2$	0.3; 0.5; 0.6; 0.8
$C{=}CH_2 \cdots C{=}C$	0.5
$CH_4 \cdots F{-}CH_3$	0.2

Fluorine substituent effect

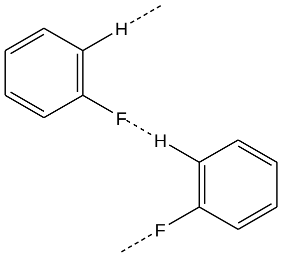
Heinrich Heine Universität Düsseldorf

On Interactions?

- C-H···F interactions similar to C-H···O und C-H···N, but very weak
- C-H···F well established but pretty unpredictable, system dependent



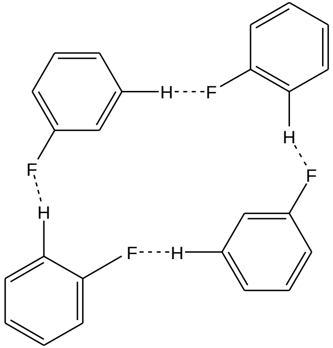
I



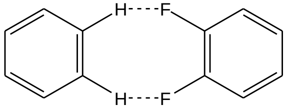
II

and still…

Classification of packing motifs with C-F···H interactions



III



IV

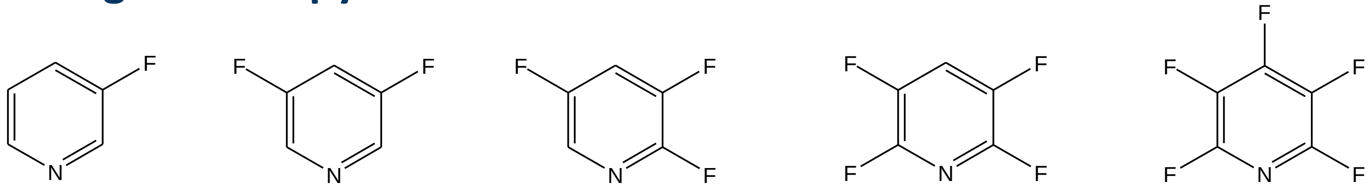
C-H···F comparable to C-H···N hydrogen bonds
➢ stabilizing specific crystal structures

V. Vasylyeva, K. Merz *J. Fluorine Chem.*, **2010**, *131*, 446

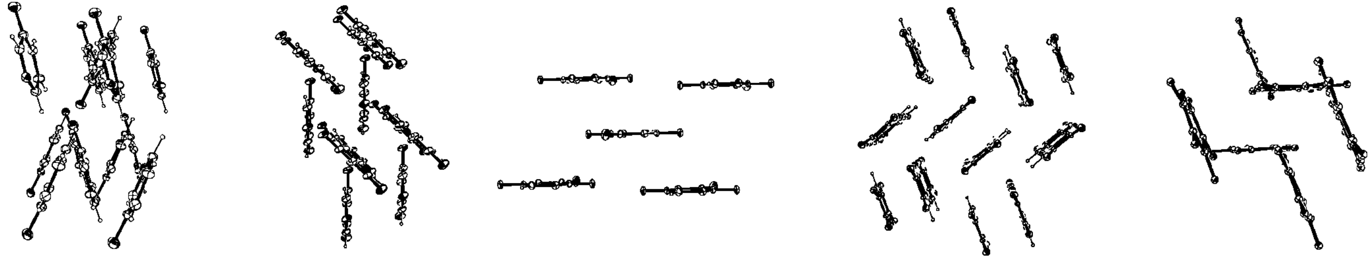
What about F···F contacts?

Heinrich Heine Universität Düsseldorf

Crystal packing of fluoropyridines



3-F-Py | 3,5-F-Py | 2,3,5-F-Py | 2,3,5,6-F-Py | penta-F-Py**



1. Planarisation with a growing number of fluorine substituents
2. Switching back to edge-to-face aggregation

V. Vasylyevaet al *Cryst. Growth Des.,* **2012**, 12, 1032; V. Vasylyeva, K. Merz, *J. Fluorine Chem.*, **2010**, *131*, 446; ** A. Olejniczak, et al, *J. Fluorine Chem.*, **2008**, *129*, 173

Questions

Crystal is a solid form of a substance with a definite and **periodic** internal structure.

I. What is the essential, **basic structural motive (BSM)** in the crystal packing?

BSM – infinite structural fragment

II. Which types of **intermolecular interactions dominate** the molecular aggregation?

Possibilities for the interpretation of the crystal packing:

- **geometrical --** visualization of the crystal packing

- energetic -- computational methods

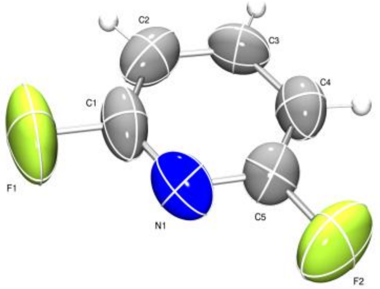
T. Steiner, *Angew. Chem. Int. Ed.* **2002**, *41*, 48

Geometrical interpretation of crystal packing

➢ What You See Is What You Want

Heinrich Heine Universität Düsseldorf

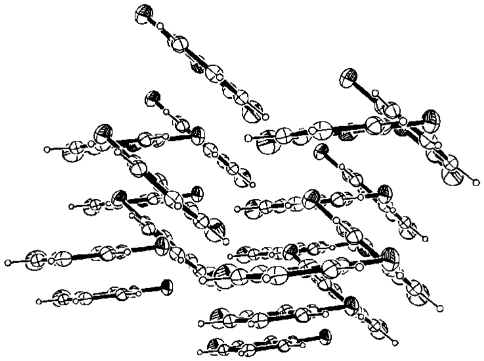
2,6-Difluoropyridine:
geometrical interpretation



- distorded edge-to-face arrangement

- inter-ring angle: 46.8°
- centroid-centroid distance: 6.90 Å

- no short F···F-, F···N-, F···H- or C···H-contacts



crystal packing overview

Energetical interpretation

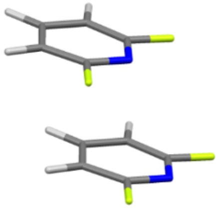
contribution of the energy of intermolecular interactions

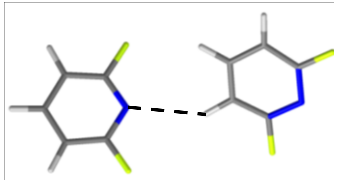
→ *ab-initio* calculations (e.g. method/basis set: **MP2/ 6-311 G**)**

→ based on experimental data (single crystal X-Ray analysis)

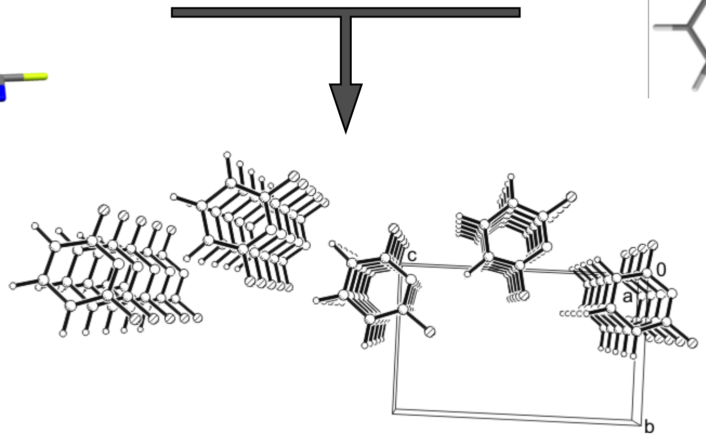
2,6-FP: Reconstructing crystal packing based on the most negative energy values

π···π'
(-2.01 kcal/mol





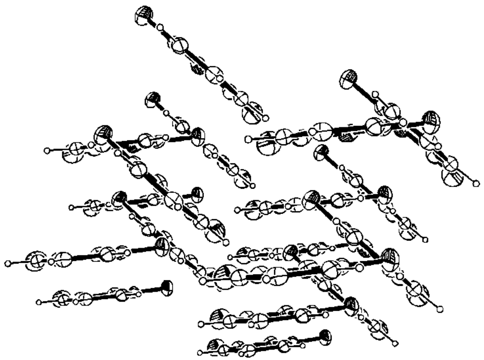
C-H···N'
(-2.44 kcal/mol)



V. Vasylyeva, O. V. Shishkin, A. V. Maleev, K. Merz, *Cryst. Growth Des.*, **2012**, *12*, 1032-1039

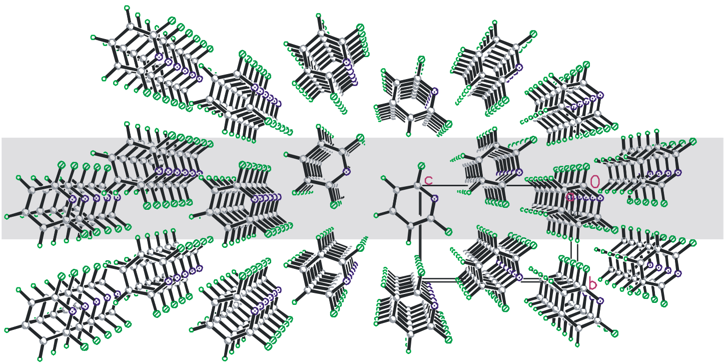
2,6-Difluoropyridine

geometrical view



edge-to-face arrangement

energetic view

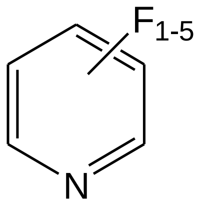


zig-zag layer

$E_{int}^{BSSEcorr}$ = - 10.67 kcal/mol

E_{ext} = - 2.11 kcal/mol

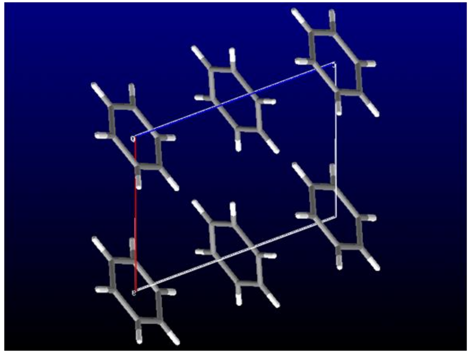
Interactions: C-H···N & stacking



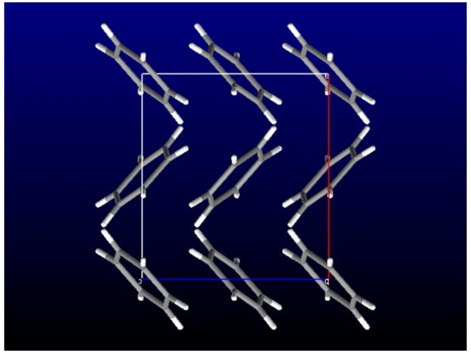
>> short F···F distances:
- smallest negative or
- even positive values

Crystal packing of benzene

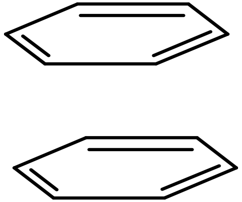
- isostructural to fluorobenezene C_6F_6



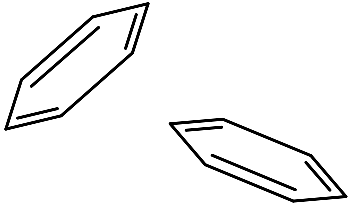
high pressure modification



modification at 1 atm



face-to-face π··π stacking

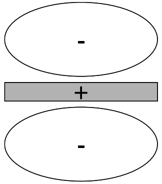


edge-to-face C-H··π interaction

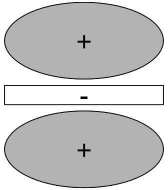
Crystal Engineering with C_6H_6 and C_6F_6

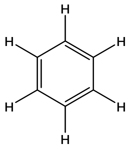
Quadrupol moments:

C_6H_6

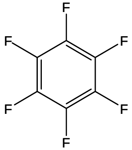


C_6F_6

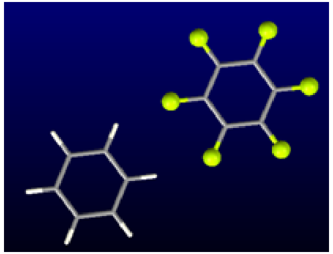




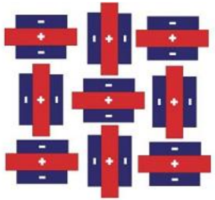
benzene



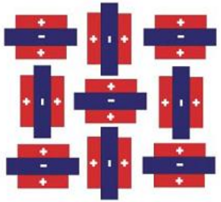
hexafluorobenzene

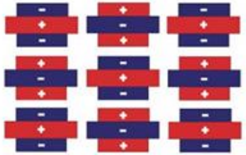


C_6H_6

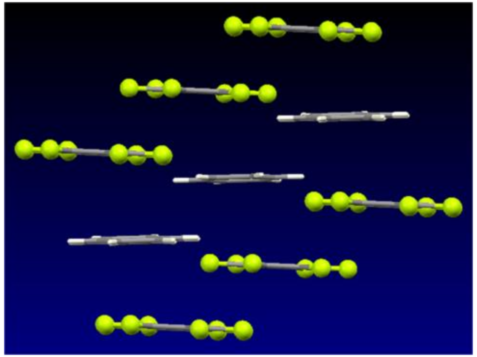


C_6F_6





C_6H_6 x C_6F_6 co-crystal



π···π stacking

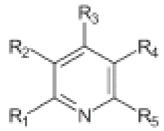
→ aryl-perfluoroaryl interaction

Aryl-perfluoroaryl interactions in CE co-crystal formation

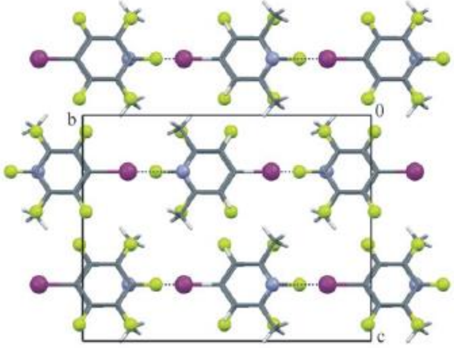
➢ XB + aryl-perfluoraryl stacking

C_6F_5I

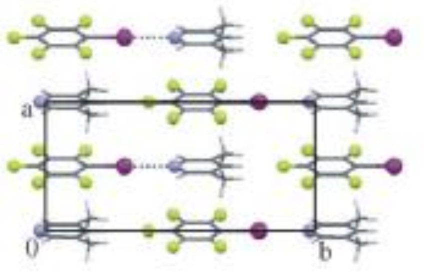
1



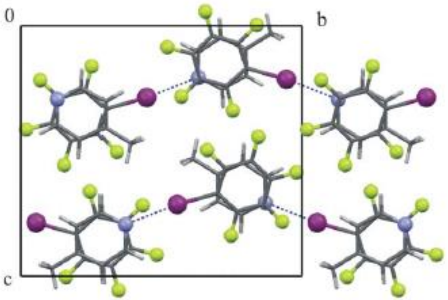
2 $R_1 = R_3 = R_5 =$ Me; $R_2 = R_4 =$ H
3 $R_1 = R_5 =$ Me; $R_2 = R_3 = R_4 =$ H
4 $R_2 = R_4 =$ Me; $R_1 = R_3 = R_5 =$ H
5 $R_3 =$ Me; $R_1 = R_2 = R_4 = R_5 =$ H



(a)



(c)



(b)

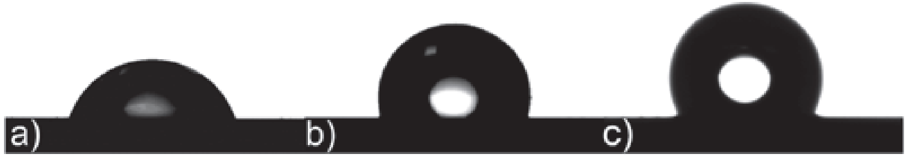
Polonski et. al, CrystEngComm, 2007, 9, 203–206

Linear and branched perfluoropolyether coatings on diamond-like carbon films: covalent linkage and physical deposition

WALTER NAVARRINI[1], MAURIZIO SANSOTERA[1]*, FRANCESCO VENTURINI[1], CLAUDIA L. BIANCHI[2], ANTONIO GUARDA[3], GIUSEPPE RESNATI[1]

Application of liquid lubricant films on diamond-like carbon (DLC) coatings.
Hard surfaces are typically lubricated with perfluoropolyethers (PFPE).

Water droplets on several DLC coatings



blank DLC thin film

linear PFPE coating

branched PFPE coating

Resnati et. al, Chemistry Today/Fluorine Chemistry, 2010

CHEMICAL REVIEWS
Review
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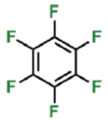
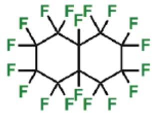
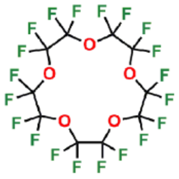
^{19}F Magnetic Resonance Imaging (MRI): From Design of Materials to Clinical Applications

Ilaria Tirotta,[†,‡] Valentina Dichiarante,[†,‡] Claudia Pigliacelli,[†,‡] Gabriella Cavallo,[†,‡] Giancarlo Terraneo,[†,‡] Francesca Baldelli Bombelli,[*,†,‡,§] Pierangelo Metrangolo,[*,†,‡,∥] and Giuseppe Resnati[*,†,‡]

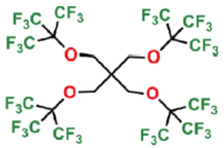

HFB


PFD

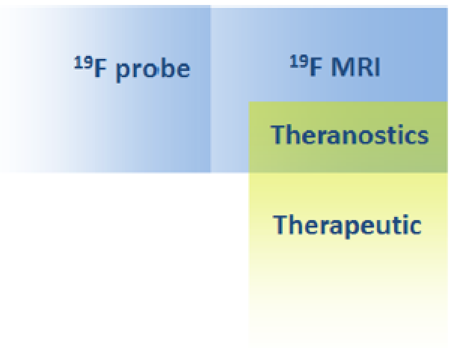

PFN


PFCE

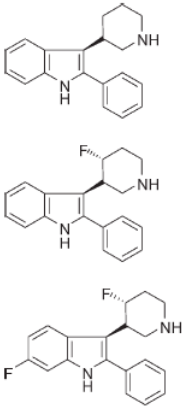

PFOB


PERFECTA

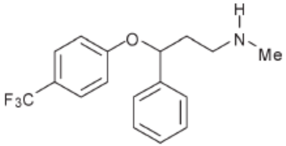
Polyfluorocarbons as MRI tracers:
molecular tracers, polymers, branched derivatives



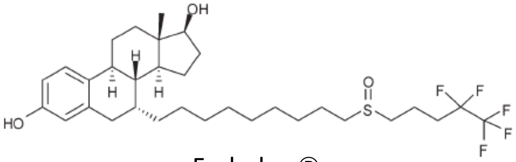
Fluorine in medicinal chemistry



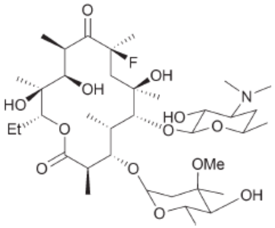
3-piperidinylindole
antipsychotic drugs



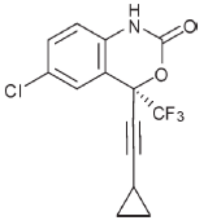
Prozac ®
antidepressant



Faslodex ®
oestrogen receptor antagonist



Flurithromicin
antibiotic



Efavirenz
transcriptase inhibitor HIV treatment

In a series of 3-piperidinylindole: fluorination decreased the basicity of the amine, thereby improving bioavailability

Supramolecular synthon

Combine knowledge about supramolecular chemistry and
apply on solid materials (molecular crystals)

crystal = supermolecule & intermol. interactions = glue

Remember CE's aim = design and control the packing of the molecules

– HOW?–

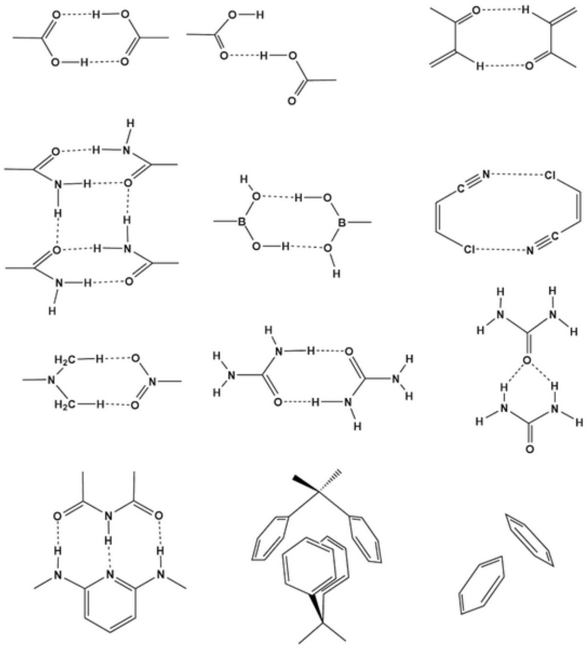
- find a recurring packing pattern
- find a reliable functional group which makes a robust motive
- use it to create a new solid-state material

Supramolecular synthon

Supramolecular synthons *are structural units within a crystal which can be formed and/or assembled by known or conceivable synthetic operations involving intermolecular interactions.*

G. R. Desiraju, Angew. *Chem. Int. Ed.* **1995**, *34*, 2311–2327.

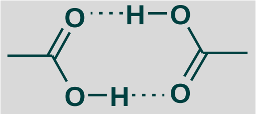
- small repetitive units
- fully describe the essence of the structure
- define specific intermolecular interactions

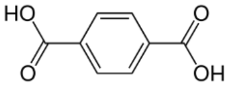


Some commonly used synthons.

Simplification of complex systems describing crystals as **networks**

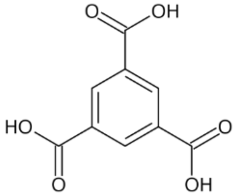
Starting from carboxyl dimer synthon: Building up dimensionality





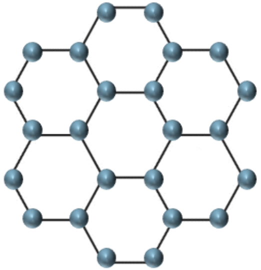
terephthylic acid

1D



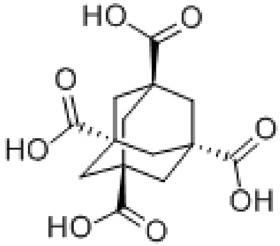
trimesic acid

2D



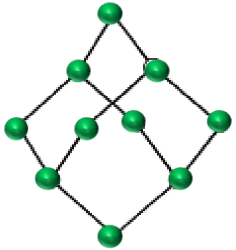
Nodes
and
Connectors

Molecules – Nodes
Synthons – Connectors



adamantane-1,3,5,7- tetracarboxylic acid

3D

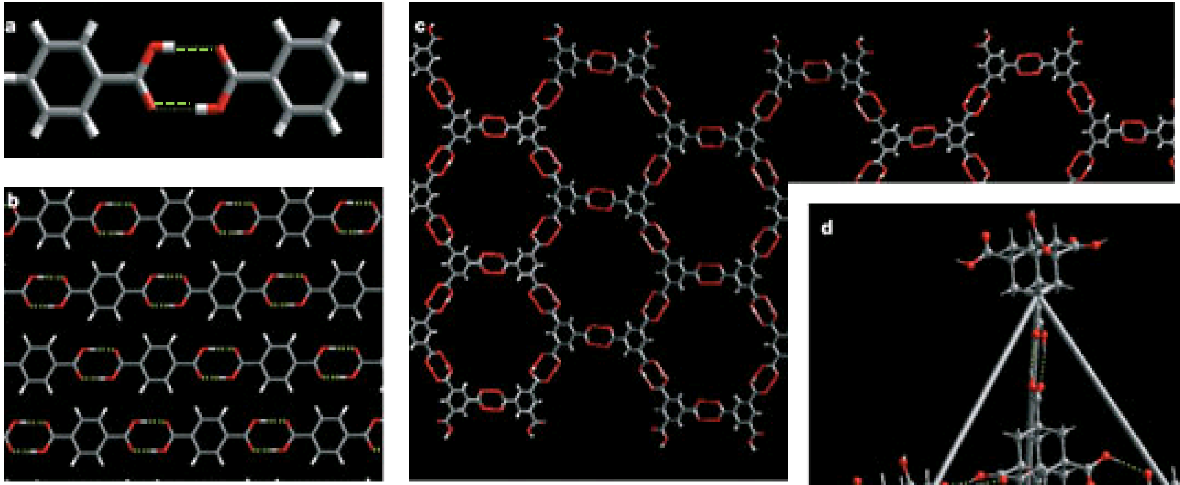


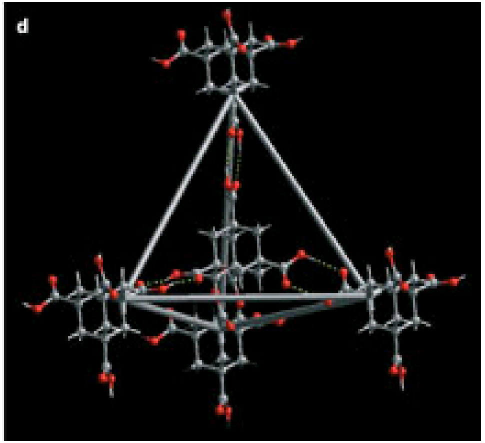
CE with carboxylic acids

master key - HB

Heinrich Heine
Universität Düsseldorf

0D entity: dimer as supramolecular synthon





1D array: linear arrangement in chains
2D layers: planar chicken-wire network
3D networks equivalent to the diamond lattice

Nodes and connectors approach

➢ Definition of a crystal structure as a network, involving nodes (molecules) and node connections (interactions)

HCN - hydrogen cyanide

- linear sequence of molecules (nodes “ • ”)
- linked with N≡C-H···N≡C-H hydrogen-bonds (node connections ” - ”)

•••N≡C-H•••N≡C-H•••N≡C-H•••N≡C-H•••

working back from the crystal structure - retro-synthesis principle:

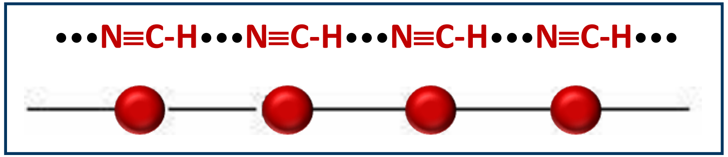
→ any bifunctional molecule with C-H and C≡N-groups (forming C-H···N≡C)
→ will take the same network structure

Nodes and connectors approach

- Definition of a crystal structure as a network, involving nodes (molecules) and node connections (interactions)

Heinrich Heine Universität Düsseldorf

H-C≡N - hydrogen cyanide

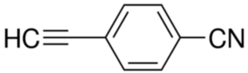


H-C≡C-C≡N - cyanoacetylene

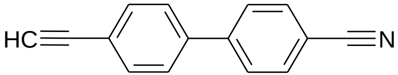
- understanding of the cyrstal structure
- assembles in the same linear array

Further: introduce a phenyl ring between the HB functionalities

4-ethynylcyanobenzene



4'-Ethynyl-4-cyanobiphenyl



C-H···N≡C hydrogen bonds is a supramolecular synthon

What is a useful supramolecular synthon?

➢ Criteria

→ **frequent occurrence in a given group of compounds**

- strength and directionality of interaction
- commonness of the functional group
- the ease with which the group can be introduced into the molecule

→ **synthon formation is more specific**

- question of probability of formation

→ **synthon formation can lead to meaningful fragments in its disconnection**

- e.g. C-H···H-C synthon: very common, but not useful for planning

→ ***combines* an economy in size with a maximum of structural information**

- small SS: very frequently - but no much specific information (O-H···O-H)
- large SS: plenty of structural information - but observed rarely, hardly accessible

NATURE VOL. 335 15 SEPTEMBER 1988 NEWS AND VIEWS 201

Crystals from first principles

A new calculation of the polymorphs of silica appears to have broken new ground in deriving crystal structure from chemical composition. But X-ray crystallographers need not worry — yet.

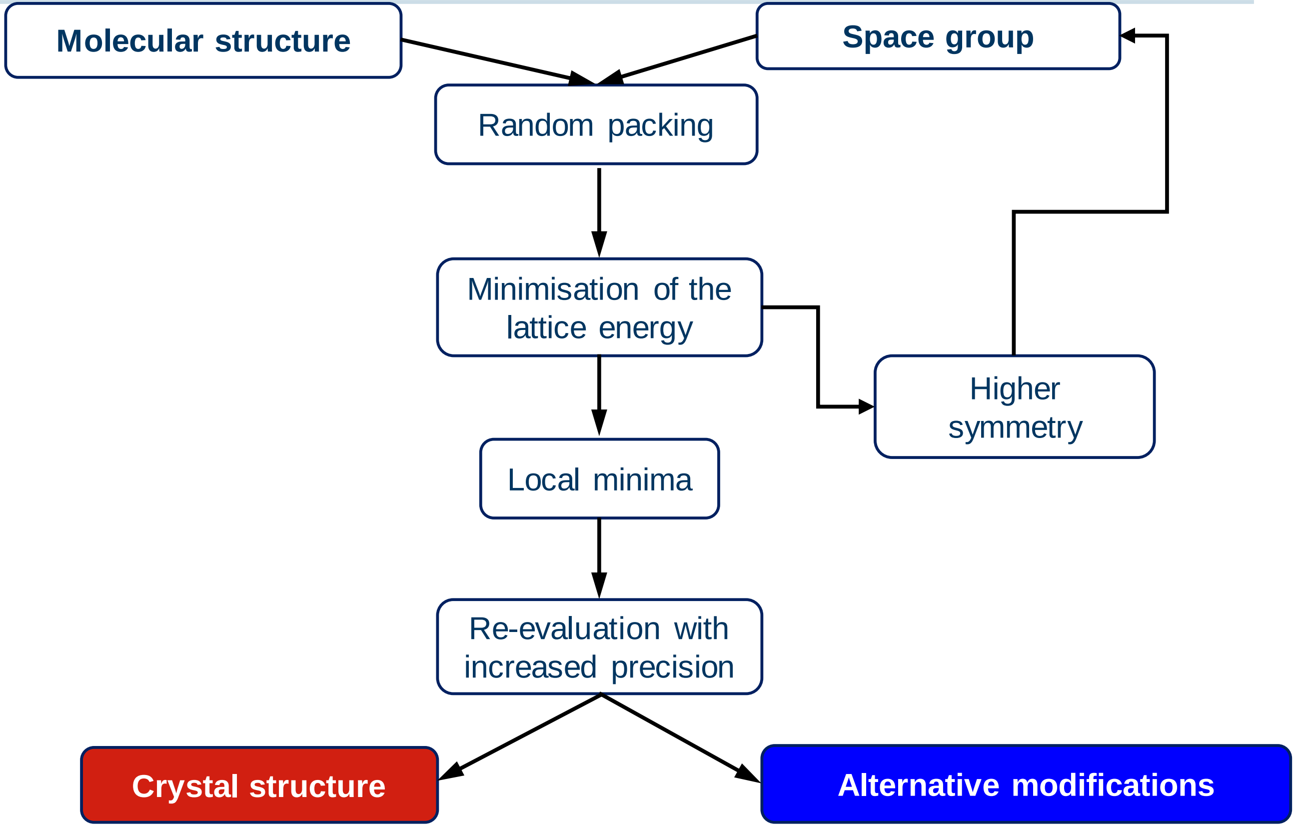
One of the continuing scandals in the physical sciences is that it remains in general impossible to predict the structure of even the simplest crystalline solids from a knowledge of their chemical composition. Who, for example, would guess that graphite, not diamond, is the thermodynamically stable allotrope of carbon at ordinary temperature and pressure? Solids such as crystalline water (ice) are still thought to lie beyond mortals' ken.

oxygen atoms — the authors estimate 0.7 of an electron charge for every SiO bond, or close on two electron charges for those O atoms bound to two Si atoms. So, to ensure that their calculated structures are electrically neutral, the authors place electronic charges at points 1.65 Å along the four tetrahedral axes of the cluster, neatly simulating in the process the electrostatic effects of the neighbouring oxygen ions found in structures such as

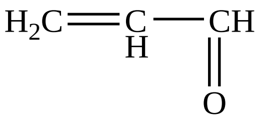
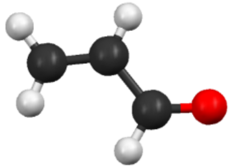
modulus within 10 per cent. The agreement between calculated energy (enthalpy) of the polymorphs and the few measurements available is even better (1 per cent), but the authors say they could probably do even better if they allowed for the band structure of these polymorphs and the extent to which the bands are filled at different temperatures.

Several refinements remain to be explored. The chances are that three-body

An Crystal Structure Prediction Approach

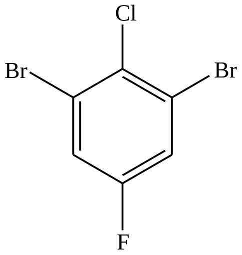


Crystal Structure Prediction (CSP)

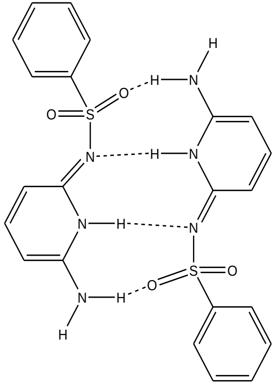
Acrolein

4th blind test, 2009
Success for small rigid molecules

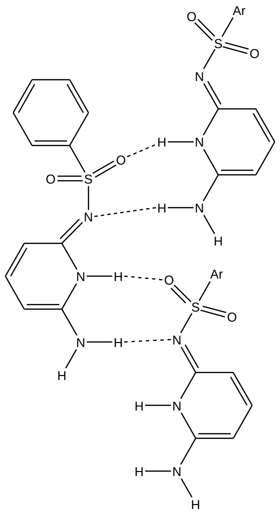


1,3-dibromo-2-chloro-5-fluorobenzene

(a)



(b)



2nd blind test, 2002
6-amino-2-phenylsulfonylimino-11,2-dihydropyridine
four-point motif (a)
two-point motif (b)

No proper prediction, because of
Flexibility & different interaction modes

6. Blind Test

➢ start 2015

(1) Rigid molecules, max. 30 atoms.

(2) Partially flexible molecule with 2–4 internal degrees of freedom, max. 40 atoms.

(3) Partially flexible molecule as a **salt** , 1-2 internal degrees of freedom, max. 40 atoms.

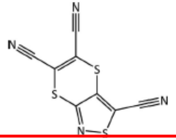
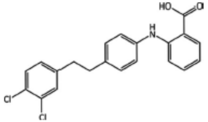
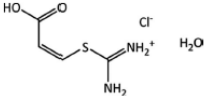
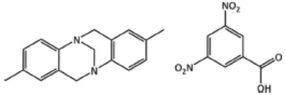
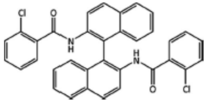
(4) Multiple partially flexible (1-2degrees of freedom) independent molecules:
→ a co-crystal or solvate; max. 40 atoms.

(5) Molecule with 4–8 internal degrees of freedom, 50–60 atoms.

→ ***Each** theoretical **research group** can suggest **3 best ranking** predictions*

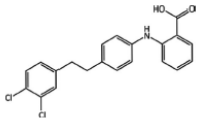
6. Blind Test

➢ published 2016, 21 groups, 90 researchers

Target	Chemical diagram	Crystallization conditions, remarks and clarifications	Attempted predictions	Times generated	Best rank (incl. re-ranking)
(XXII)		Crystallized from an acetone/water mixture; chiral-like character due to potential flexibility of the six-membered ring, but no chiral precursors used in synthesis.	21	12	1
(XXIII)		Five known polymorphs (*A*–*E*); three $Z' = 1$ (*A*, *B*, *D*), two $Z' = 2$ (*C* and *E*). The most stable polymorphs at 257 and 293 K are both $Z' = 1$. Crystallization conditions include slow evaporation of acetone solution and of ethyl acetate: water mixture.	*A*, *B* and *D*: 14; *C* and *E*: 3	*A*: 4, *B*: 8, *C*: 1, *D*: 3, *E*: 0	*A*: 23, *B*: 1, *C*: 6, *D*: 2, *E*: –
(XXIV)		Crystallized from 1 *M* HCl solution. The substituents of the C═C double bond are in the *cis* configuration.	8	1	2
(XXV)		Slow evaporation of a methanol solution, which contained a racemic mixture of the enantiomers of Tröger's base.	14	5	1
(XXVI)		Slow evaporation from 1:1 mixture of hexane and dichloromethane. No chiral precursors used in synthesis.	12	3	1

6. Blind Test 2015

- a difficult target

(XXIII)		Five known polymorphs (*A*–*E*); three $Z' = 1$ (*A*, *B*, *D*), two $Z' = 2$ (*C* and *E*). The most stable polymorphs at 257 and 293 K are both $Z' = 1$. Crystallization conditions include slow evaporation of acetone solution and of ethyl acetate:water mixture.	*A*, *B* and *D*: 14; *C* and *E*: 3	*A*: 4, *B*: 8, *C*: 1, *D*: 3, *E*: 0	*A*: 23, *B*: 1, *C*: 6, *D*: 2, *E*: –

Molecular conformations found in forms *A*–*D* of (XXIII), overlaid onto the fenemate group of the molecule; form *A* is in blue, form *B* in grey, form *C* molecule 1 is in red, form C molecule 2 in purple and form *D* in orange. H atoms are omitted for clarity.

→ ***slight differences between the polymorphs***

→ ***conformational changes***

→ ***flexibility of the phenyl group***